Rank and the Determinant

ECON 441: Introduction to Mathematical Economics
Lecture 6

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Rank of a Matrix

Rank of a matrix = maximum number of linearly independent rows

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Rank of A? Rank of B?

Full rank $\iff$ Nonsingularity

Checking for Nonsingularity

Rank of a matrix = maximum number of linearly independent rows or (equivalently) columns

If a square matrix has full rank, it is nonsingular.

To check for nonsingularity, calculate the **determinant**: for singular matrices, the determinant is zero.

Determinant

Determinant $|A|$ is a unique scalar associated with a **square** matrix A.

Determinant of a 2×2 Matrix:

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

Can be calculated as:

$$|A| = a_{11}a_{22} - a_{12}a_{21}$$

Determinant of a 3×3 Matrix

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Determinant of a $n \times n$ Matrix

A **minor** of the element a_{ij}, denoted by $|M_{ij}|$ is obtained by deleting the ith row and jth column.

Cofactor C_{ij} is defined as:

$$|C_{ij}| = (-1)^{i+j}|M_{ij}|$$

Then,

$$|A| = \sum_{i=1}^{n} a_{ij}|C_{ij}| = \sum_{j=1}^{n} a_{ij}|C_{ij}|$$

Find the Determinant

$$A = \begin{bmatrix} 1 & 5 & 1 \\ 0 & 3 & 9 \\ -1 & 0 & 0 \end{bmatrix}$$

Properties of Determinants

$|A| = |A^T|$

Interchanging rows or columns will alter the sign but not the value

Multiplication of any one row (or one column) by a scalar k will change the value of the determinant k-fold

The addition (subtraction) of a multiple of any row (or column) to (from) another row (or column) will leave the determinant unaltered

If one row (or column) is a multiple of another row (or column), the value of the determinant will be zero.

Criteria for Nonsingularity

The following statements are equivalent:

$|A| \neq 0$

Rows (or equivalently columns) of A are independent

A has full rank

A is nonsingular

A^{-1} exists

A unique solution to $Ax = b$ ($x^* = A^{-1}b$) exists

Homework Questions

- Exercise 5.2: 1 (c), (e), (f), 2, 3, 6
- Exercise 5.3: 1, 4, 5, 8

Textbook reference: 5.1-5.3